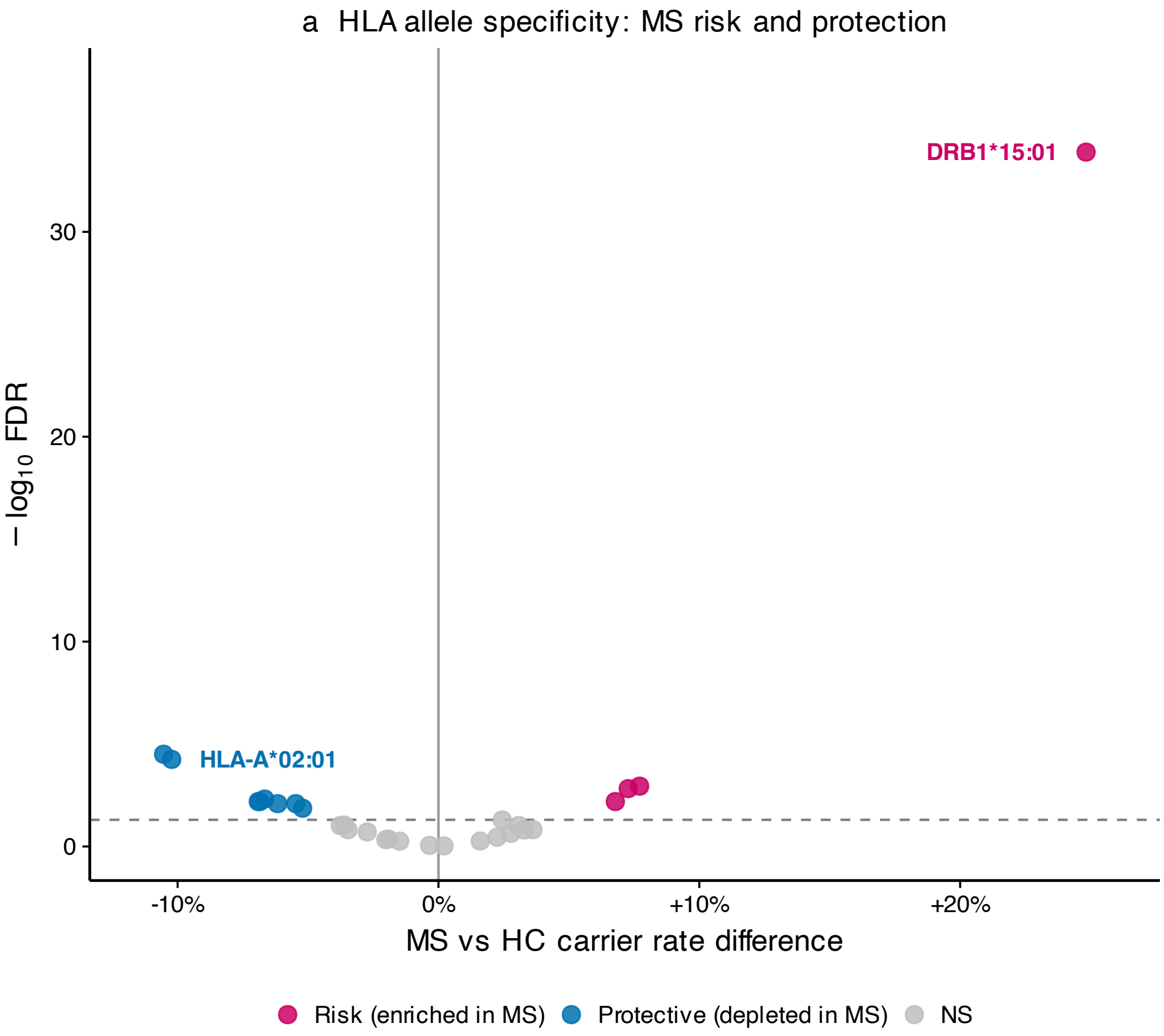
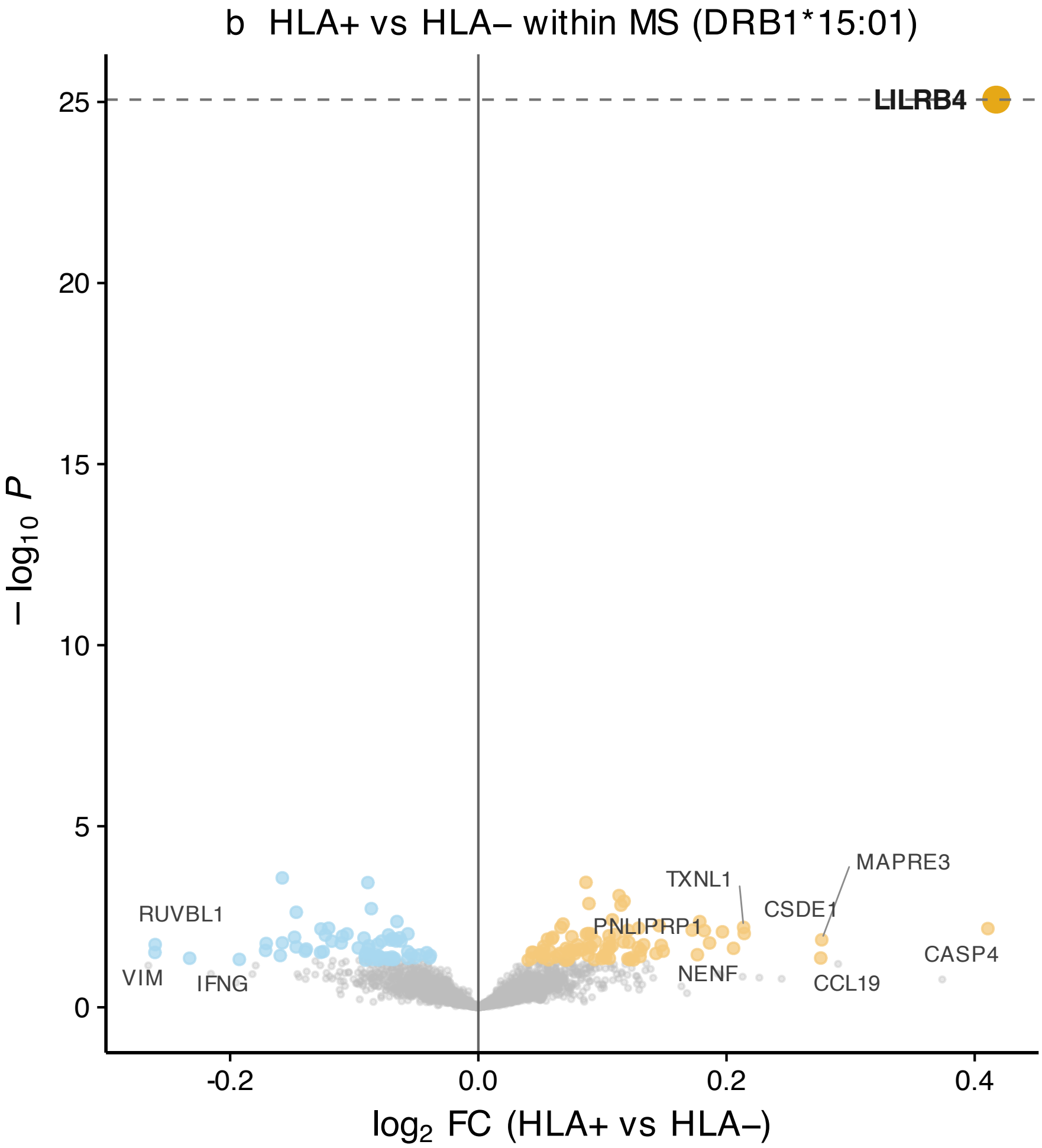
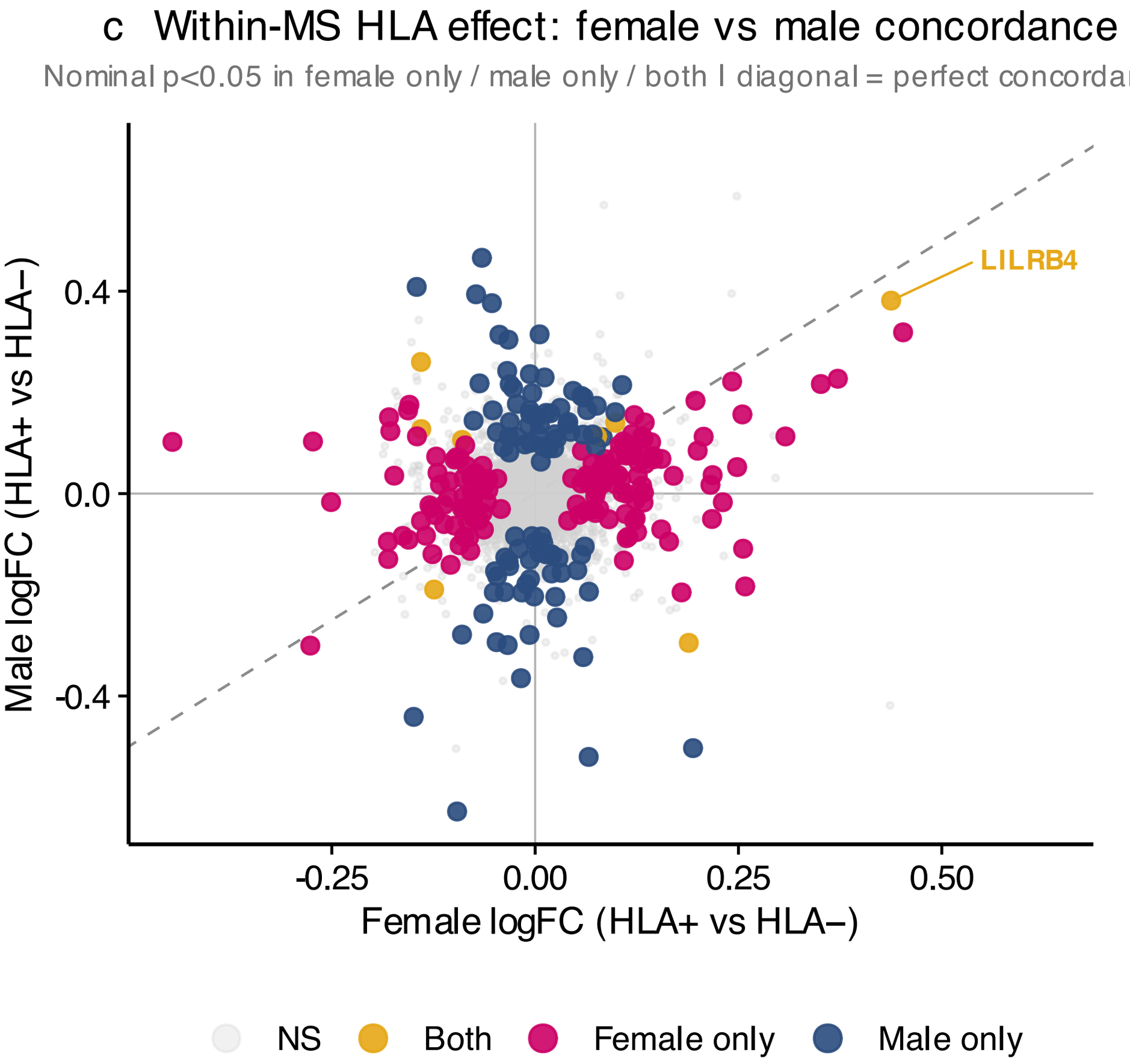
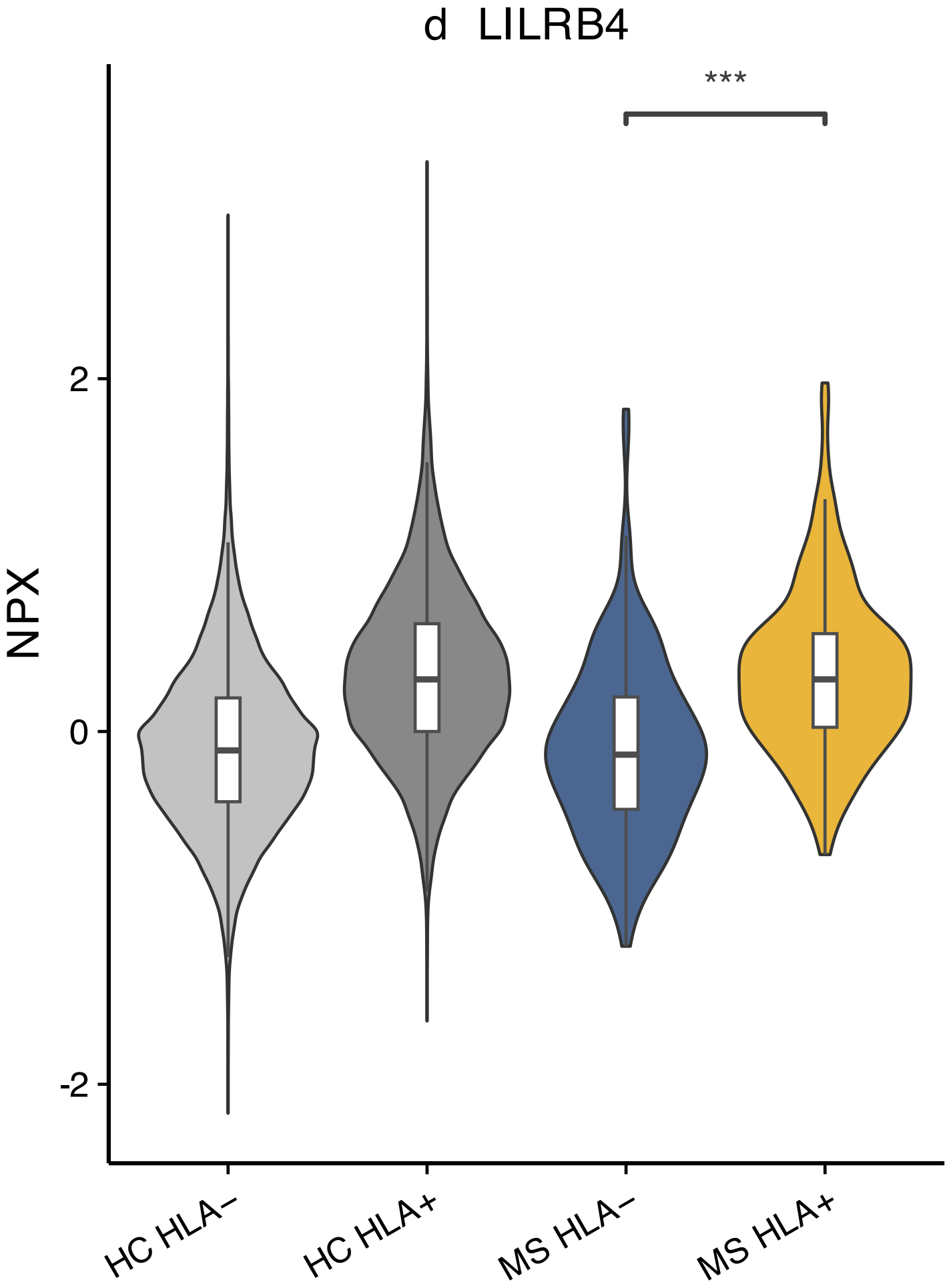
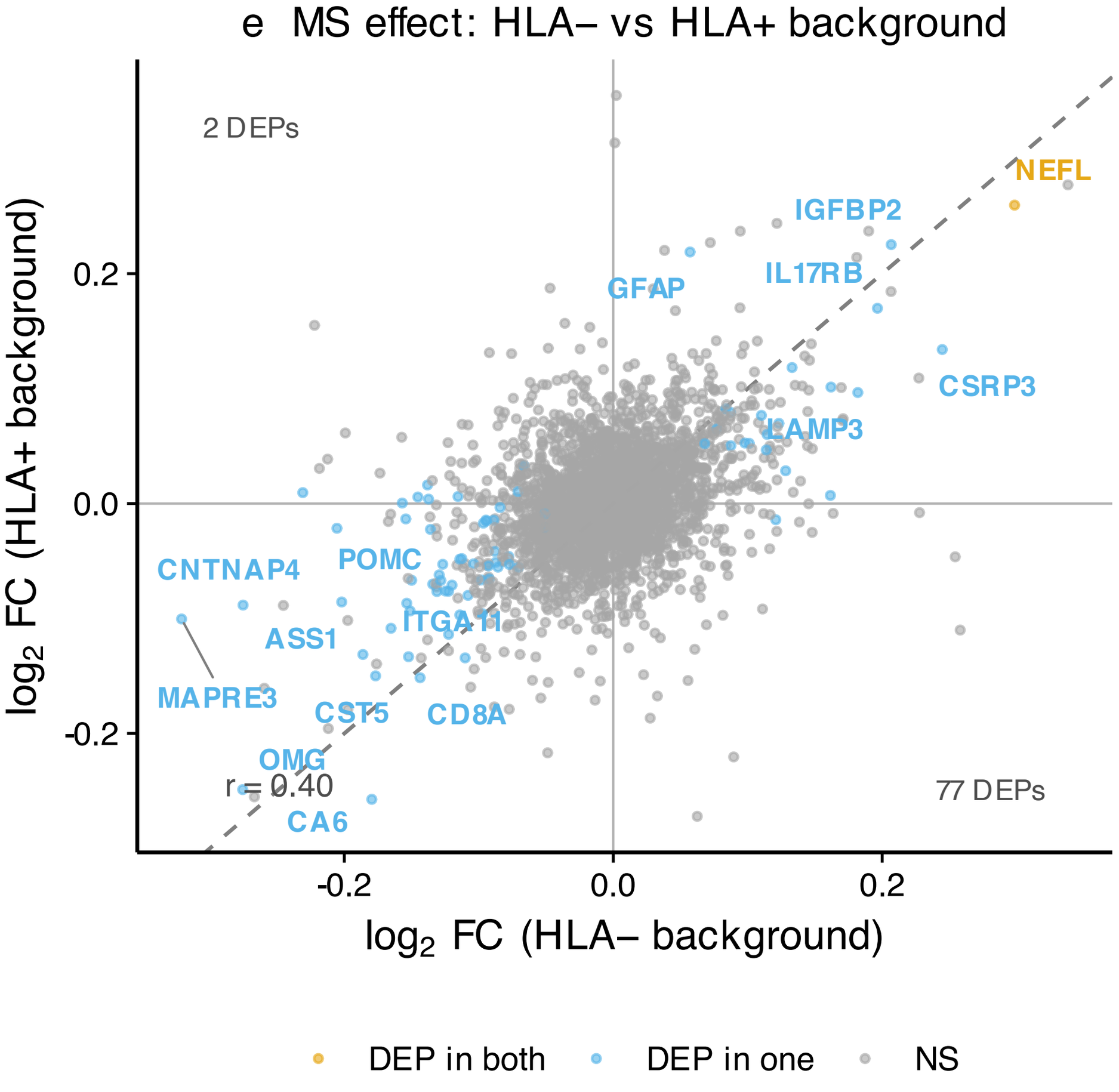
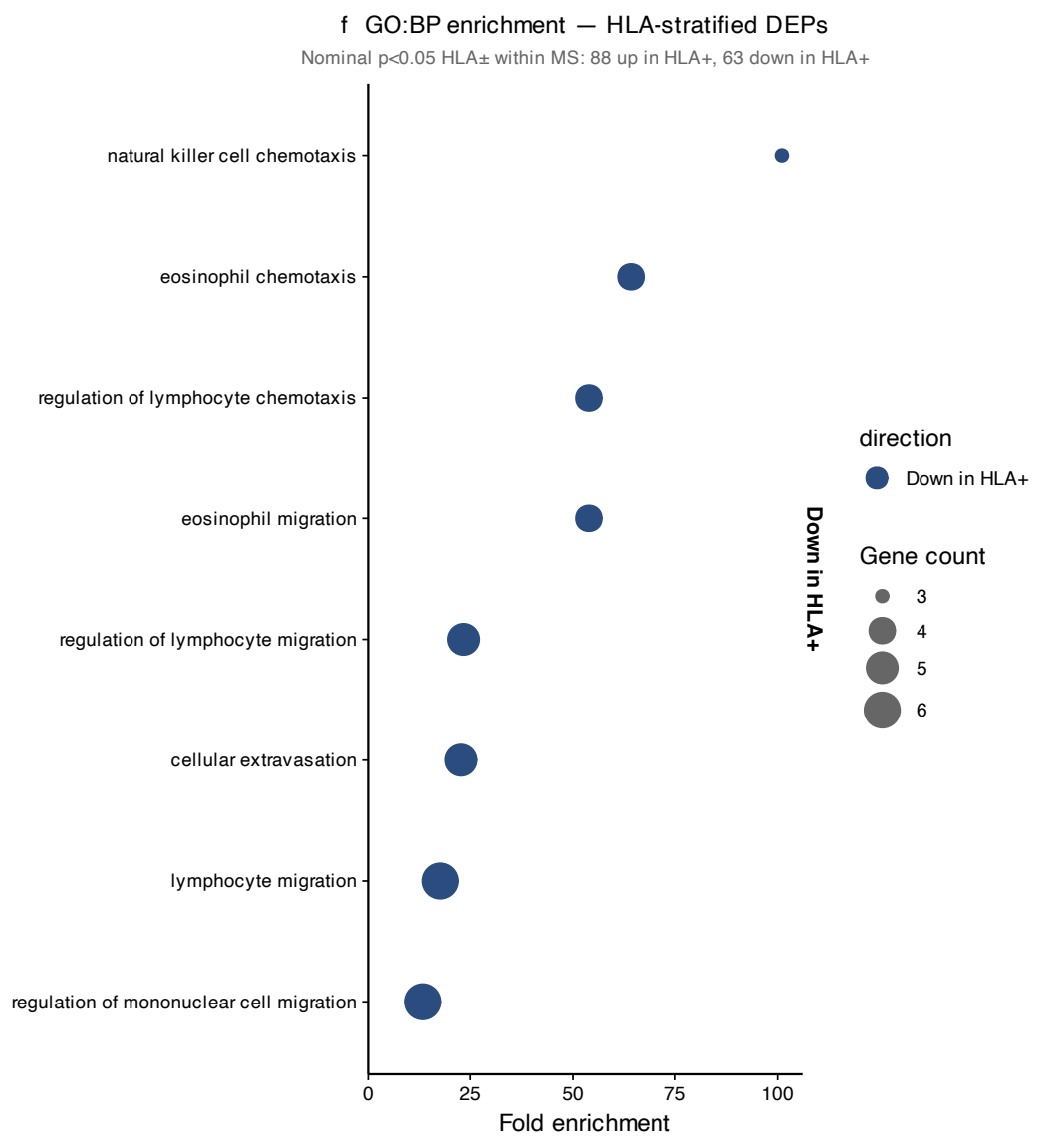

a HLA allele specificity: MS risk and protection
DRB1*15:01
HLA-A*02:01
30
20
10
0
− log10 FDR
-10%
0%
+10%
+20%
MS vs HC carrier rate difference
Risk (enriched in MS)
Protective (depleted in MS)
NS

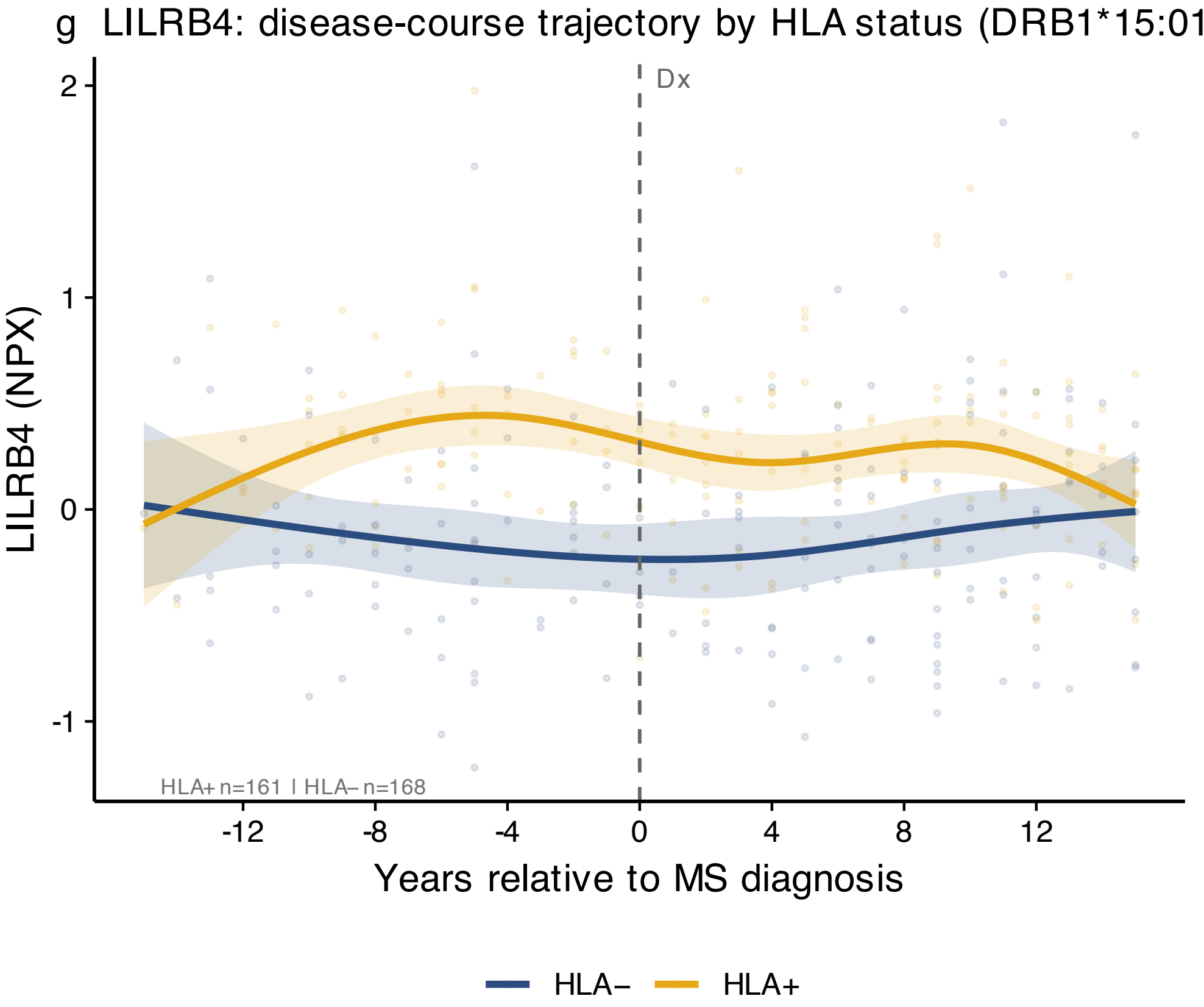

b HLA+ vs HLA– within MS (DRB1*15:01)
LILRB4
MAPRE3
TXNL1
CSDE1
RUVBL1
PNLIPRP1
CASP4
VIM
IFNG
NENF
CCL19
25
20
15
10
5
0
-0.2
0.0
0.2
0.4
$-\log_{10} P$
$\log_2$ FC (HLA+ vs HLA–)

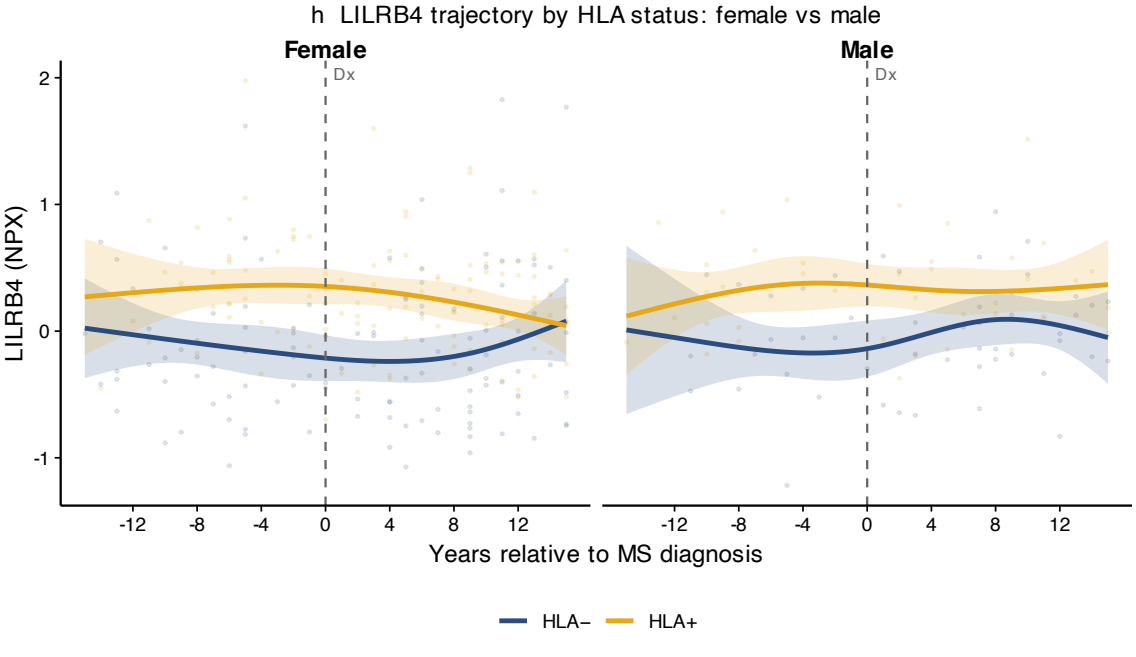

c Within-MS HLA effect: female vs male concordance
Nominal p<0.05 in female only / male only / both | diagonal = perfect concordan
LILRB4
0.4
0.0
-0.4
Male logFC (HLA+ vs HLA−)
-0.25
0.00
0.25
0.50
Female logFC (HLA+ vs HLA−)
NS
Both
Female only
Male only

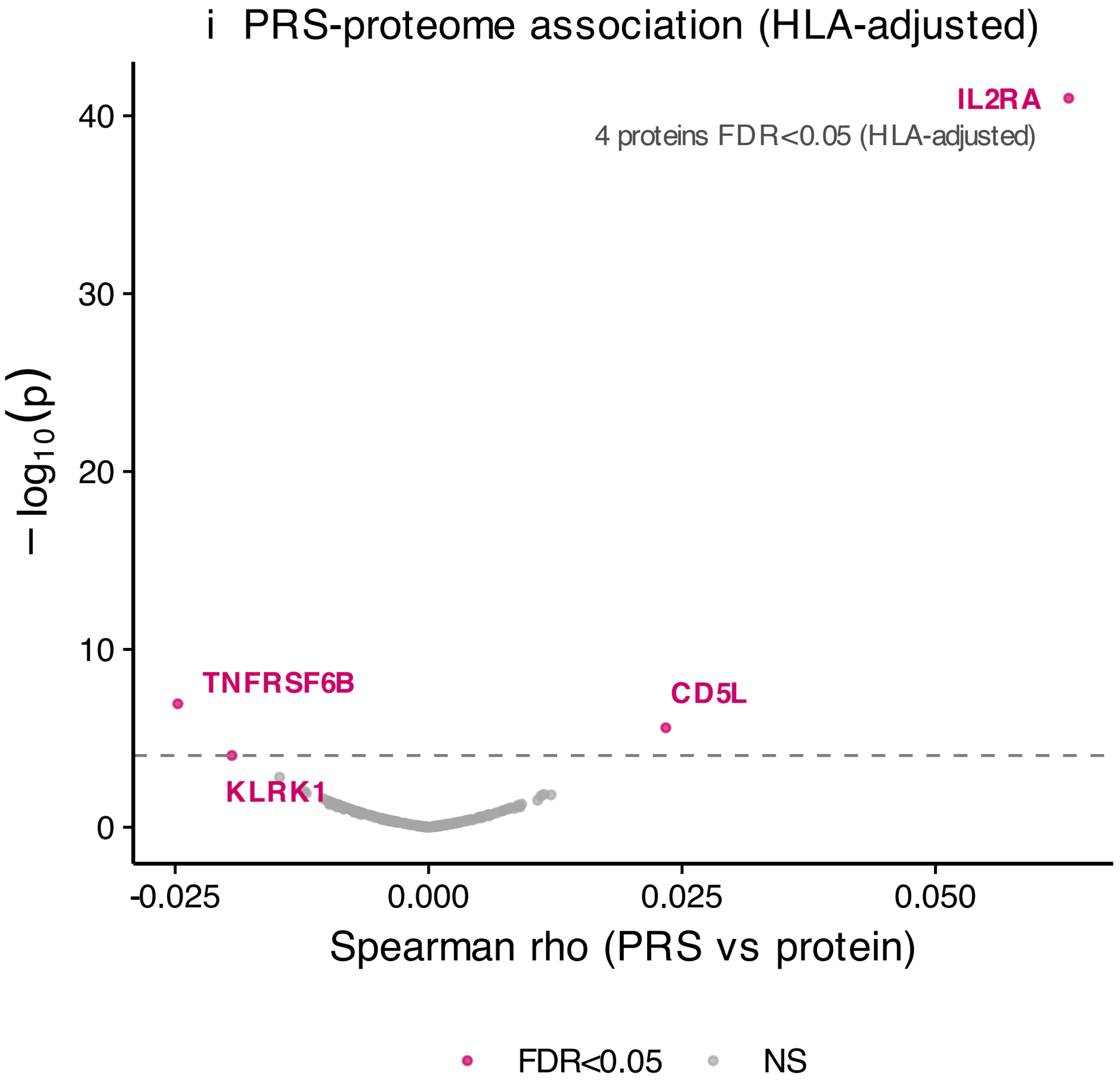

d LILRB4
***
NPX
2
0
-2
HC HLA−
HC HLA+
MS HLA−
MS HLA+

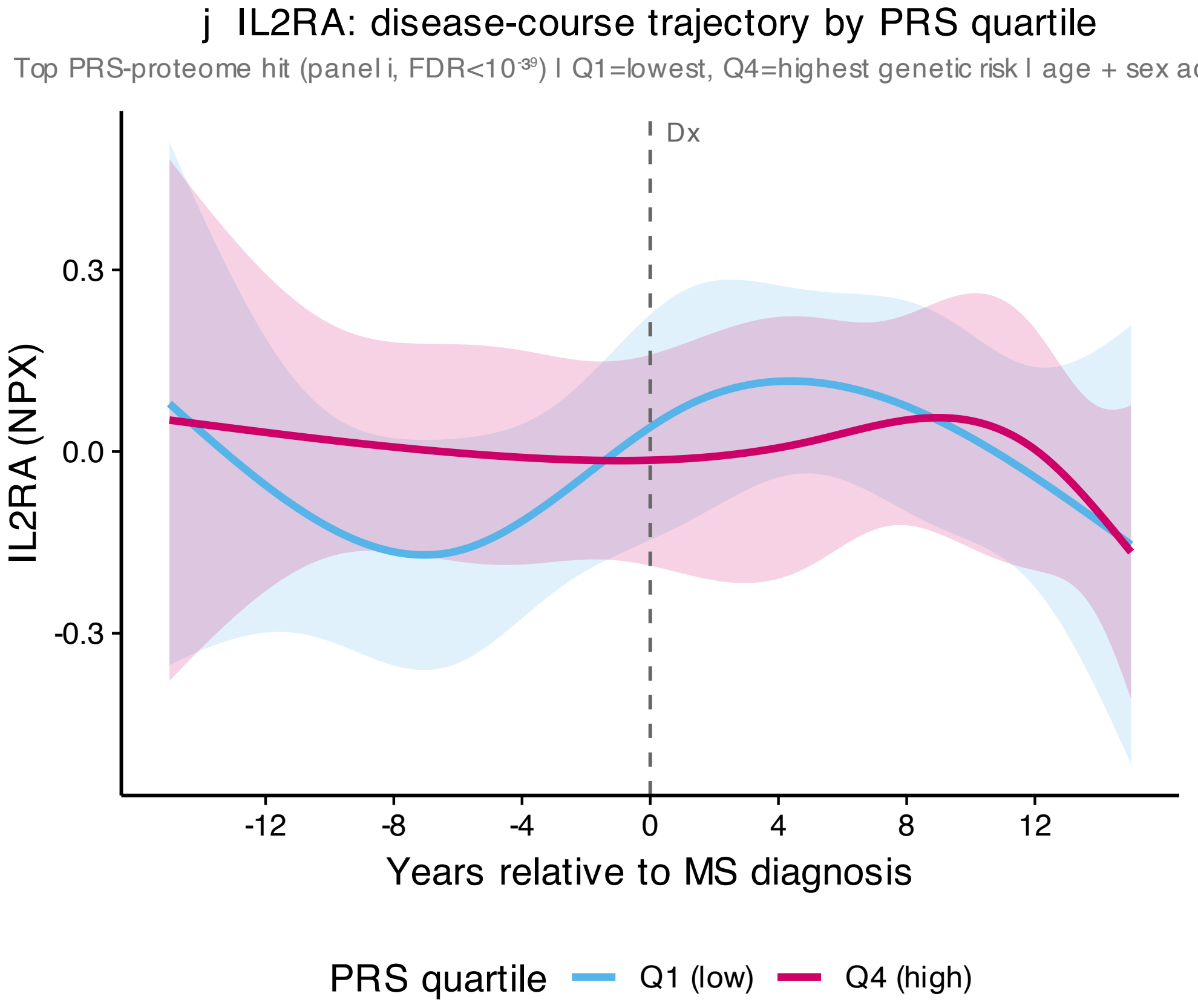

e MS effect: HLA– vs HLA+ background
2 DEPs
77 DEPs
NEFL
IGFBP2
IL17RB
GFAP
CSRP3
LAMP3
CNTNAP4
POMC
ITGA11
ASS1
MAPRE3
CST5
CD8A
OMG
CA6
r = 0.40
0.2
0.0
-0.2
-0.2
0.0
0.2
log2 FC (HLA+ background)
log2 FC (HLA– background)
DEP in both
DEP in one
NS

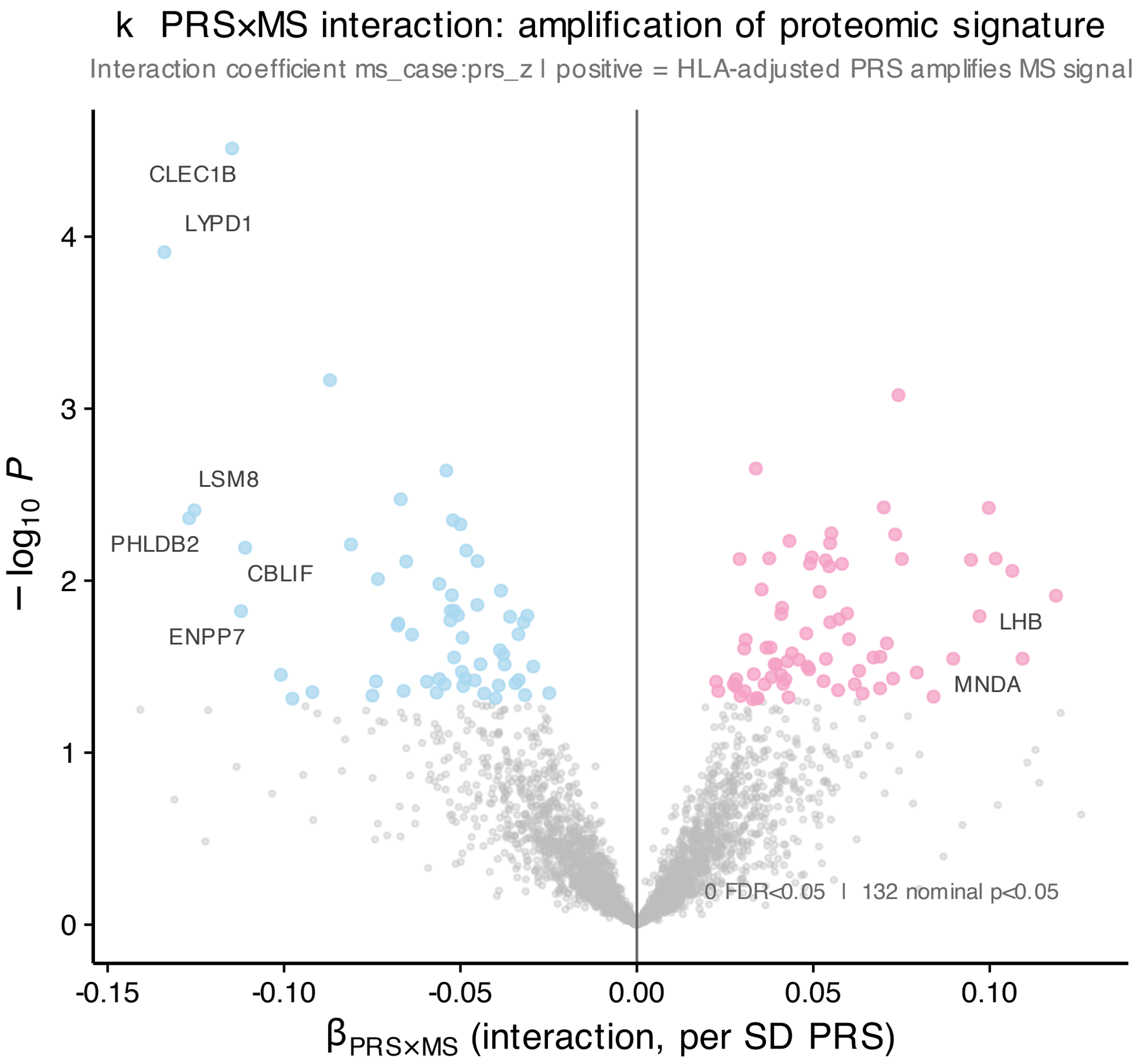

f GO:BP enrichment — HLA-stratified DEPs
Nominal p<0.05 HLA± within MS: 88 up in HLA+, 63 down in HLA+
natural killer cell chemotaxis
eosinophil chemotaxis
regulation of lymphocyte chemotaxis
eosinophil migration
regulation of lymphocyte migration
cellular extravasation
lymphocyte migration
regulation of mononuclear cell migration
Down in HLA+
0
25
50
75
100
Fold enrichment
direction
Down in HLA+
Gene count
3
4
5
6


g LILRB4: disease-course trajectory by HLA status (DRB1*15:01
Dx
LILRB4 (NPX)
2
1
0
-1
HLA+ n=161 | HLA− n=168
-12
-8
-4
0
4
8
12
Years relative to MS diagnosis
HLA−
HLA+


h LILRB4 trajectory by HLA status: female vs male
Female
Male
Dx
Dx
LILRB4 (NPX)
2
1
0
-1
-12
-8
-4
0
4
8
12
Years relative to MS diagnosis
HLA−
HLA+


i PRS-proteome association (HLA-adjusted)
IL2RA
4 proteins FDR<0.05 (HLA-adjusted)
TNFRSF6B
CD5L
KLRK1
40
30
20
10
0
−log10(p)
-0.025
0.000
0.025
0.050
Spearman rho (PRS vs protein)
FDR<0.05
NS


j IL2RA: disease-course trajectory by PRS quartile
Top PRS-proteome hit (panel i, FDR<10-39) | Q1=lowest, Q4=highest genetic risk | age + sex ad
Dx
0.3
0.0
-0.3
IL2RA (NPX)
-12
-8
-4
0
4
8
12
Years relative to MS diagnosis
PRS quartile
Q1 (low)
Q4 (high)


k PRS×MS interaction: amplification of proteomic signature
Interaction coefficient ms_case:prs_z | positive = HLA-adjusted PRS amplifies MS signal
CLEC1B
LYPD1
LSM8
PHLDB2
CBLIF
ENPP7
LHB
MNDA
0 FDR<0.05 | 132 nominal p<0.05
−log10 P
βPRS×MS (interaction, per SD PRS)
-0.15
-0.10
-0.05
0.00
0.05
0.10
0
1
2
3
4